

Electrochemistry

ELECTRICITY IS USED IN CHEMISTRY TO ALTER COMPOUNDS OR TRANSFER MATERIALS.

SEE ALSO

◀ 112–113 Ionic bonding

◀ 133 Electrochemistry

Refining metals

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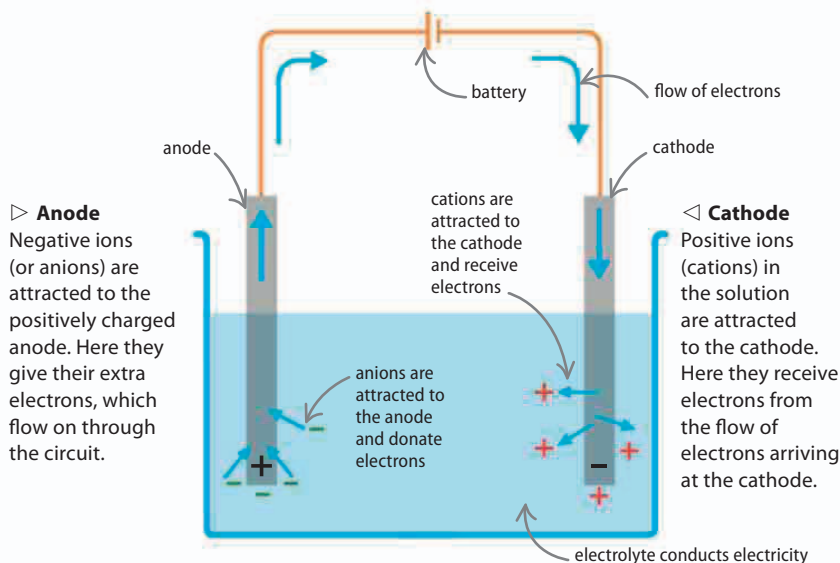
Electricity

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The energy carried by electric currents can be used in chemistry. Currents are frequently used to force compounds apart, by converting ions back into atoms to produce pure elements.

Electrolytes

An electrolyte is a liquid that conducts electricity. It is an ionic compound and has to be liquid (molten or in a solution) so that its component ions are free to move. A power source is connected to two electrodes that are placed in the electrolyte. This creates a positive charge at one electrode (the anode) and a negative charge at the other (the cathode). The positive and negative ions in the electrolyte then move toward the electrode with the opposite charge, where they receive or donate electrons. This flow of ions carries electricity through the liquid.



Electrolysis

Passing an electric current through an ionic compound will split it into its component elements. This is called electrolysis and was the process used to isolate many new elements for the first time. When the power source is turned on, the positive and negative ions in the compound are attracted to their oppositely charged electrodes. At the cathode, positive ions receive electrons and, at the anode, negative ions lose electrons, so the ions become neutral atoms again. The pure elements build up at each electrode and can be collected. Water is an ionic compound (H_2O) that can be split into hydrogen (H) and oxygen (O) in this way.

